

UZH451 - Spacecraft Communications

Term Project

1. Form groups of two or three. You will design a static link, and analyze the corresponding link budget.
2. Use the "Satellite Link Budget Analyzer" in MATLAB (on the Apps tab, you may need to install the required "Satellite Communications" Toolbox firstly).
3. After opening "Satellite Link Budget Analyzer", select the "Static Link Budget Analyzer".
4. By default, a scenario of three links is given. Simply, deactivate the satellite-to-satellite link. You will have one downlink and one uplink with respect to the ground station.
5. Select a geostationary satellite, and two distinct ground stations. You will build a communication link between two ground stations through the satellite. The first ground station will be the broadcasting entity (uplink from GS1 to Satellite). The second ground station will be a regular satellite dish receiver (downlink from Satellite to GS2).
6. Search for the required parameters including geographic coordinates, receiver and transmitter properties, antenna properties and link properties. Provide the datasheets, and cite the resources that you used. If you have difficulty in finding the parameters, select some commercial antennas and equipment that will fit your scenario, and provide datasheets and resources.
7. After entering the parameters in the fields in the "Link Canvas", hit the "Analyze" button (do not tick the "Include P.618 Losses checkbox"). Then, the link budget will be provided to you.
8. See the following Link Design Procedure given in the "Chapter 4 Satellite Link Design" of Pratt's book. Check if the Link Budget results match your calculations.
9. Use the "Add Plot" button to create your "2-D Contour Plots". Generate at least 5 contour plots, and make your comments on them by explaining X, Y and Z axes clearly.
10. If you enjoy the tool, go deeper with "Time Varying Link Budget Analyzer" to build your own scenarios with satellite TLEs to analyze link budgets in the given time window. (Not necessary for this term project, just for your curiosity.)
11. Submit a single zip file including your term project file and datasheets if you have. Your term project should include a list of resources which are accordingly cited within the text.
12. Use a Google account to use the link to enter your group.

<https://docs.google.com/spreadsheets/d/1hmPp-0rSCn55Y-yYVgAvqMynV8ruRb0bVvcvCmqaNBI/edit?usp=sharing>

Satellite Communication Link Design Procedure

The design procedure for a one-way satellite communication link can be summarized by the following 10 steps. The return link design follows the same procedure.

1. Determine the frequency band in which the system must operate. Comparative designs may be required to help make the selection.
2. Determine the communications parameters of the satellite. Estimate any values that are not known.
3. Determine the parameters of the transmitting and receiving earth stations.
4. Start at the transmitting earth station. Establish an uplink budget and a transponder noise power budget to find $(C/N)_{up}$ in the transponder.
5. Find the output power of the transponder based on transponder gain or output backoff.
6. Establish a downlink power and noise budget for the receiving earth station. Calculate $(C/N)_{dn}$ and $(C/N)_0$ for a station at the edge of the coverage zone (worst case).
7. Calculate S/N or BER in the baseband channel. Find the link margins.
8. Evaluate the result and compare with the specification requirements. Change parameters of the system as required to obtain acceptable $(C/N)_0$ or S/N or BER values. This may require several trial designs.
9. Determine the propagation conditions under which the link must operate. Calculate outage times for the uplinks and downlinks.
10. Redesign the system by changing some parameters if the link margins are inadequate. Check that all parameters are reasonable, and that the design can be implemented within the expected budget.